

Towards Multimodal User State–Aware Adaptive Guidance in MR

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ABSTRACT

Most mixed reality (MR) guidance systems rely on static instructions, overlooking user behavioral and cognitive variability. We present a multimodal adaptive guidance framework that combines LLM-generated task graphs with semantic step tracking driven by multimodal interaction events. The system also continuously logs gaze, hand, and head signals to support future temporal user state modeling. In an example MR guidance task, the system achieves robust step tracking with real-time responsiveness (≈ 60 ms action-to-classification latency). Together, these results demonstrate reliable task progress monitoring and establish a foundation for user state–aware adaptive MR guidance.

Index Terms: Mixed Reality, Multi-modal Interaction, Large Language Model, Adaptive Guidance, User State Estimation.

1 INTRODUCTION

Mixed reality (MR) training systems typically rely on static one-size-fits-all instructions, limiting their ability to adapt to users with diverse prior knowledge, goals, and needs. As a result, existing approaches struggle to respond when users hesitate, deviate from the intended procedure, or require additional explanation. Adaptive guidance therefore requires structured task representations, real-time monitoring of user actions and cognitive–emotional signals, and flexible control over feedback timing and granularity [3]. We present a work-in-progress adaptive MR framework toward user state–aware guidance that combines large language model (LLM)–based task-graph construction, semantic step tracking, and real-time multimodal logging. This poster evaluates the feasibility and real-time performance of core components, while user state modeling and closed-loop adaptation are left for future work. Specifically, we contribute (1) an LLM-generated task-graph representation for MR task-oriented guidance, (2) a semantic action recognition and step-tracking pipeline grounded in multimodal interaction logs, and (3) a real-time logging infrastructure enabling future user state–aware adaptation. We report system performance across six MR sessions ($F1 = 0.97$, ≈ 60 ms action-to-classification latency), demonstrating robust semantic step tracking even under off-protocol behavior.

2 PROPOSED FRAMEWORK

The framework comprises four components: task-graph construction and step tracking, multimodal sensing and feature extraction, temporal modeling, and an LLM-based adaptive decision logic. The current implementation realizes task-graph, step-tracking, and sensing pipelines, with temporal modeling and adaptation planned for future integration (see Fig. 1).

Graph Builder and Step Tracker: Task progress is modeled as a directed acyclic graph (DAG) of instructional steps automatically generated from natural-language protocols via LLM

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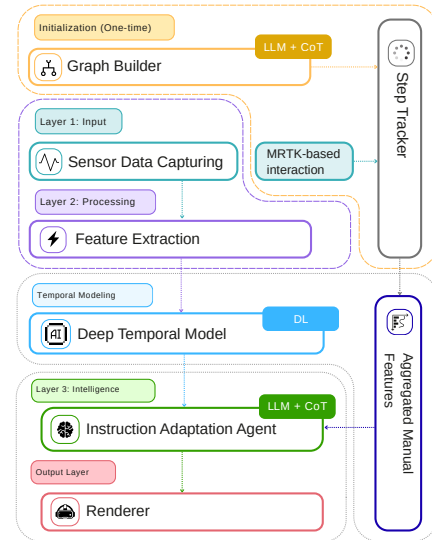


Figure 1: Multimodal adaptive guidance framework

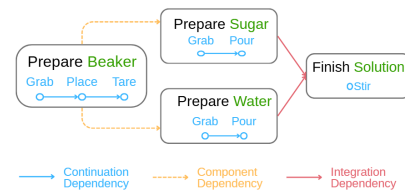


Figure 2: Example step-tracking DAG for a sugar solution task. Nodes denote instructional steps; blue arrows indicate within-step continuation, dashed orange arrows denote component dependencies, and red arrows mark integration points where branches merge.

prompting and manually verified before evaluation. Each node represents a canonical semantic action enriched with metadata such as dependencies, alternative execution paths, and error severity, while edges encode ordering constraints (see Fig. 2). This structure enables step-level progress tracking and mistake classification inspired by the AMMA framework [4]. Each high-level step may include optional substeps for future adaptive feedback generation. On the **sensing layer**, Microsoft’s MRTK interaction logging is combined with object-aware context from a YOLO-based detection pipeline on HoloLens 2. Low-level hand (grab, release, poke) and gaze (enter, exit, dwell) events for detected objects are streamed in real time to a Python backend via a lightweight UDP bridge. A **heuristic action recognition** layer maps short temporal patterns of these interaction events to interpretable action primitives (e.g., placing, pouring, pressing). For instance, a sequence of grabbing, moving and releasing an object is interpreted as a *place* action. Finally, the **step tracker** matches these actions with the LLM-generated DAG, considering dependency constraints and alternative paths and classifies each into AMMA-style outcome categories

(e.g., expected, premature, repeat, recoverable error). The system maintains the current task state and available next steps, generating concise feedback for the user interface and statistical summaries that are intended to feed aggregated manual features for future adaptive guidance. This modular pipeline enables reusable, interpretable task monitoring while reducing reliance on fixed scripts.

Multimodal Sensing and Feature Extraction: We log multimodal signals on HoloLens 2 using MRTK3 as a basis for future cognitive and affective state estimation. The sensing setup captures gaze direction signals and dwell-related events, hand joint trajectories, and head pose at runtime. Due to different tracking rates, all modalities are temporally synchronized and recorded into a unified sensor log at a fixed rate of 30 Hz, providing a consistent time base for downstream temporal modeling. From these logs, we extract a compact, interpretable feature set intended for sequence-based state estimation, including hand, gaze, head-motion, and interaction-context features. The resulting representation can be aggregated into temporal windows for downstream modeling. A detailed list of extracted features is provided in the supplemental materials.

Temporal Modeling and Aggregated Features: Prior work shows that cognitive load in MR can be modeled using either feature-based or end-to-end deep learning approaches [1, 2]. Our framework supports flexible **temporal modeling**: raw 30 Hz streams can feed end-to-end Transformers, while extracted features feed lightweight classifiers or sequence models, balancing interpretability, data efficiency, and real-time feasibility. In addition to sensor features, we compute aggregated, **interpretable task metrics** (e.g., cumulative error and repetition counts from the step tracker) that can later be passed to the guidance adaptation module to support explainable user state-aware feedback. Temporal modeling is under active development; this poster focuses on sensing and feature extraction that will support future integration of user state estimation into adaptive guidance.

Adaptive Decision Logic: To personalize guidance, the LLM-based instruction adaptation module is designed to maintain evolving user profiles (novice, intermediate, expert) and control guidance granularity, feedback modality, and tone. These profiles are updated via prompts conditioned on step-tracker signals and, in future work, estimated cognitive and emotional state. For instance, novice prompts provide step-by-step explanations with visual cues, whereas expert prompts offer concise feedback unless repeated errors or uncertainty are detected, and this adaptive decision logic is being implemented as part of a closed-loop guidance architecture.

3 PRELIMINARY RESULTS

We conducted six Microsoft HoloLens 2 sessions with a single expert user: two followed the intended protocol (clean), two included predefined injected errors (error), and two allowed free-form exploratory behavior (exploratory). Across all sessions, semantic action recognition achieved an F1 of 0.97 ± 0.03 , outcome classification - 0.96 ± 0.05 , and step completion detection - 1.00, indicating robust protocol tracking even under off-protocol behavior. Deviation detection latency on erroneous or off-path actions was higher and more variable (77 ± 65 ms) due to the additional reasoning required, but still suitable for real-time corrective feedback **Tab. 1**. The latency from physical action to classified outcome remained low across all session types (clean: 62 ± 18 ms, error: 62 ± 7 ms, exploratory: 51 ± 4 ms), indicating responsive real-time feedback **Fig. 3a**. Per-session results show high performance in clean sessions (action/outcome F1=1.00/1.00), and high robustness in error sessions (0.94/0.93) and exploratory sessions (0.97/0.96), suggesting that the heuristic action layer and DAG-based outcome classifier correctly handle actions that deviate from the primary task sequence without sacrificing responsiveness. The aggregated outcome-class

Table 1: System performance metrics by session type. Action and outcome denote F1 scores for semantic action recognition and outcome classification; dev. lat. indicates deviation detection latency.

Category	Action, F1	Outcome, F1	Dev. Lat. (ms)
Clean (n=2)	1.00	1.00	116 ± 89
Error (n=2)	0.94	0.93	69 ± 45
Exploratory (n=2)	0.97	0.96	45 ± 29
Mean (n=6)	0.97	0.96	77 ± 65

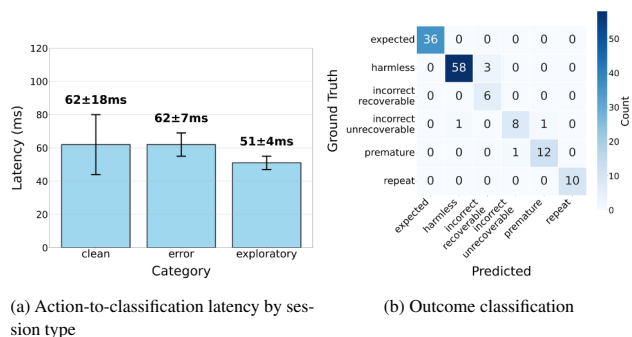


Figure 3: Preliminary evaluation results across session categories.

confusion matrix **Fig. 3b** shows clear separation for expected, premature, and repeat steps, with most remaining confusions occurring between harmless and incorrect-but-recoverable actions, reflecting borderline cases rather than failures in protocol tracking. These trials serve as a proxy for future chemistry-lab procedures such as solution preparation **Fig. 2**, allowing us to validate tracking and latency before deployment.

4 CONCLUSION AND FUTURE WORK

The presented results indicate that LLM-generated task graphs combined with semantic step tracking and MRTK interaction logs enable robust, low latency monitoring of task progress in MR (F1 = 0.97, ≤ 62 ms), including off-protocol user behavior. The framework establishes an interpretable foundation for real-time, flexible task monitoring from multimodal interaction events. However, the current evaluation is limited to small-scale sessions (n = 6) and focuses on system-level performance rather than user outcomes or subjective experience, with temporal user state modeling and closed-loop adaptive guidance under active development. Future work will incorporate temporal state estimation to enable dynamic guidance strategies.

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